



Epigenetic targeting DNMT1 of pancreatic ductal adenocarcinoma using interstitial control release biodegrading polymer reduced tumor growth through hedgehog pathway inhibition

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ABSTRACT

Annually, 48,000 people die from pancreatic ductal adenocarcinoma (PDAC), ranking it the fourth among cancer-related deaths in the United States. Currently, anti-cancer drugs are not effective against PDAC, and only extends survival by 3 months. Aberrant DNA methylation has been shown to play an important role during carcinogenesis in PDAC, with approximately 80% of tumor overexpressing the DNA methyltransferase 1 (DNMT1) protein. In the present study, we used DNMTs as a screening platform to find a new DNMT inhibitor, n-butyridenephthalide (n-BP), which is identified from a Chinese herbal drug. n-BP could inhibit DNMT1 expression in both dose-dependent and time-dependent manner. It also displays an effect in suppressing growth of PDAC cells and inducing cell cycle arrest at G0/G1 phase leading apoptosis. Growth suppression can be restored by the overexpression of DNMT1 in PDAC cells. Furthermore, we found n-BP-mediated DNMT1 suppression influenced the protein stability rather than changing the RNA expression. Through microarray studies, we found that the patched domain contained 4 (PTCHD4) is the potential downstream gene of DNMT1. Following silencing of *PTCHD4* expression by siRNA, n-BP decreased tumor growth inhibition. Finally, *in vivo*, two animal models were used to evaluate the efficacy and survival after n-BP treatment by interstitial control release polymer delivery. The results show that n-BP could effectively inhibit PDAC tumor volume growth and extend animal survival. In summary, n-BP may inhibit the growth of human PDAC cells through reducing DNMT1 and increasing the expression of PTCHD4 both *in vitro* and *in vivo*.

1. Introduction

Pancreatic ductal adenocarcinoma (PDAC) is a lethal disease because it is highly invasive and chemo-resistant. In the United States, patients diagnosed with PDAC are about 51,000, and out of that statistic, 48,000 of those patients die within a year, and the 5-year survival

rate among patients is less than < 8% [1]. The current standard treatment for PDAC is surgical resection and chemotherapy. Surgery is the only potentially curative treatment for pancreatic cancer at present. However, due to specific tumor locations, there are limitations in surgery, with only 20% of patients having the opportunity to receive surgical treatment [2]. Chemotherapy is recommended for metastatic

Abbreviations: n-BP, n-Butyridenephthalide; BP-Wafer, p(CPPSA) polymers containing BP; CHX, Cycloheximide; CPP, 1,3-bis (p-carboxyphenoxy) propane; DNMT, DNA methyltransferase; FDA, Food and Drug Administration; GLI1, GLI Family Zinc Finger 1; IC, Inhibitory concentration; IVIS, *in vivo* imaging system; LINE1, Long interspersed nuclear elements 1; MTT, 3-(4,5-Dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide; PARP, Poly (ADP-ribose) polymerase; PDAC, Pancreatic ductal adenocarcinoma; PED, paclitaxel eluting device; PTCHD4, patched domain contained 4; SA, Sebacic Acid; SAH, S-denosylhomocysteine; SHH, Sonic hedgehog; SMO, Smoothed

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pancreatic cancer. Currently, Gemcitabine agent-based chemotherapy is considered the standard of care for patients. Despite the median survival time, Gemcitabine is increased by only 1.2 months compared to fluorouracil alone [3].

Unusual epigenetic modifications involving both DNA methylation and histones modification have demonstrated a relationship between tumor initiation and progression [4,5]. DNA methylation, a covalent chemical modification resulting in addition of a methyl group at position 5 of cytosine mainly in a CpG dinucleotide, is catalyzed by DNA methyltransferases (DNMTs). In comparison with normal cells, human cancer cells exhibit hypermethylation of tumor-suppressor genes which mediate silencing of many genes, including cell cycle regulation, DNA repair, and apoptosis [6–8]. The function of DNMT1, a dominant form of the DNMTs in humans is maintaining the DNA methylation pattern after DNA replication. On the other hand, DNMT3 is responsible for DNA methylation de novo. The level of expression of DNMTs proteins has been reported to often raise a variety of cancers, especially DNMT1 [9]. In nearly 40% of patients with pancreatic cancer, DNMT1 has been overexpressed [10]. The inhibition of DNMT1 via RNA interference have demonstrated that would block the hypermethylation of newly synthesized DNA strands and lead to re-expression of silenced genes [11]. Some compounds like 5-aza-2'-deoxycytidine have been shown can inhibit DNMT1 expression, induce cancer cell apoptosis, and reduce tumor growth [12,13]. There is a high potential for the development of this group of inhibitors for cancer treatment. 5-aza-2'-deoxycytidine has therapeutic efficacy in acute myeloid leukemia and myelodysplastic syndrome in clinical. However, it has strong side effects and toxicity [14,15]. Therefore, development of new DNMTs inhibitors to replace 5-aza-2'-deoxycytidine is an important interest.

The Hedgehog pathway play an important role during embryonic development [16]. Hedgehog pathway activation requires hedgehog ligands, SHH to bind to the hedgehog receptor Patched (PTCH), a 12 transmembrane domains protein. SHH binding to PTCH inhibits the repression of Smoothened (SMO). Activated SMO subsequently induce GLI1's translocation, which enters the nucleus from the cytoplasm and begins to transcribe its target gene [16,17]. Hedgehog pathway is also strongly associated with the PDAC formation. The over activation of hedgehog pathway triggering the tumorigenesis and pancreatic stellate cells fibrosis. Hedgehog ligand was found that it abnormally expressed from pancreatic intraepithelial neoplasia to invasive tumor rather than normal pancreatic tissues [18]. Moreover, pancreatic cancer stem cells also show upregulation of SHH [19]. SHH and Gli1 protein abundance in resected PDAC specimen were reported to be associated with a poor prognosis [20]. Furthermore, inhibition of Hedgehog pathway could reduce tumor proliferation as well as enhances chemotherapy effect [18,21]. It showed Hedgehog pathway as relevant therapeutic targets for both early and advanced stages of PDAC.

In this study, we explore a novel molecule of the epigenetic regulator, n-BP, in the inhibition of DNMTs. We found that the n-BP could dramatically suppresses DNMT1 protein level through promotes the degradation of DNMT1 protein. Additionally, we used an oligodeoxynucleotide-based microarray technique and found that suppression of DNMT1 could inhibit the Hedgehog pathway activation and reduce tumor growth. In the *in vivo* experiment, we developed a local delivery system, BP-wafer that enhanced transport of chemotherapeutics into the tumor. The system reduce 58% tumor size compared with the control group. Our results represent n-BP as a novel DNMT1 inhibitor for treating PDAC.

2. Material and method

2.1. Cell lines and cell culture

The human pancreas adenocarcinoma cell line PANC-1, MIA PaCa-2, BxPC3, and HPAC was obtained from Bioresources Collection and Research Center (BCRC, HsinChu, Taiwan). Cells were routinely

cultured in DMEM or RPMI1640 medium supplemented with 10% fetal bovine serum in a 5% CO₂ at 37°C incubator. DMEM medium, RPMI 1640 medium, and fetal bovine serum were purchased from Gibco (Logan, UT, USA).

2.2. Chemical

n-Butylidenephthalide (n-BP) was purchased from Alfa Aesar Inc. (MA, USA). Cycloheximide, MG132, and were purchased from Merck Inc. (Darmstadt, Germany). Dimethylsulfoxide, 3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide (MTT), and β -actin were purchased from Sigma-Aldrich (St. Louis, MO, USA). Lipofectamine 2000 and Geneticin® (G418) were purchased from Invitrogen (Carlsbad, CA, USA). DNMT1, DNMT3A, DNMT3B, PARP and P84 antibody were purchased from Genetex (Irvine, CA, USA); Caspase 3 were purchased from Cell Signaling (Beverly, MA, USA). PTCHD4, SMO and Ki67 were purchased from Abcam (Cambridge, UK). The Horseradish peroxidase linked IgG secondary antibody was purchased from Jackson ImmunoResearch (West Grove, PA, USA).

2.3. Growth inhibition assay

The viability of the cells after treated with various chemicals was evaluated using MTT assay. Cells were seeded at a density of 5000 to 8000 cells/well in 96 well plates and grown overnight. Cells were subject to various n-BP concentration treatments. After indicated period, the medium were removed and 200 μ L of 3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide (MTT) solution (500 μ g/mL) were added to each well and incubated further for 1 h. Upon termination, the supernatant was aspirated and the MTT formazan formed by metabolically viable cells was dissolved in 100 μ L DMSO. The plates were mixed for 30 min on a shaker and absorbance was measured at 595 nm using an ELISA reader (Multiskan GO, Thermo Scientific, USA). Data represent the means $\pm$ SD from three independent experiments.

2.4. RNA extraction and qRT-PCR

Total RNA were isolated by RNeasy Midi kit (QIAGEN, Germany) according to the manufacturer's protocols. One microgram of total RNA from each sample was used to generate cDNA using the Omniscript RT kit (QIAGEN, Germany) according to manufacturer's protocol. The cDNA was amplified in the presence of one Mole of the following primers: *DNMT1*: (F) 5'-GAGCCACAGATGCTGACAAA-3' and (R) 5'-CAAGTTGAGGCCAGAAGGAG-3'; *PTCHD4*: (F) 5'-TTATGGAACGTCGACATGGA-3' and (R) 5'-TGCTCTTGCAAGGAGCTTTT-3'; *GAPDH*: (F) 5'-TGAAGTGCGGAGTCAACGGATTGGT and (R) 5'-CATGTGGGCCATGAGGTCCACCAC-3'. The *GAPDH* was used as internal control. Using the quantitative RT-PCR detection system QuantStudio 3 (Thermo Scientific, CA, USA), each gene was amplified and was analyzed in triplicate with FastStart Universal SYBR Green Master (Roche, Basel, Switzerland). Relative gene expression was calculated using the $2^{-\Delta\Delta C_t}$ method using *GAPDH* for normalization.

2.5. Small interfering RNA transfected

To silence endogenous *PTCHD4*, small interfering RNA (siRNA) oligo-targeting *PTCHD4* was purchased from Santa Cruz (sc-95224, CA, USA). Cells were grown to 60% confluence in six-well plates and transfected with *PTCHD4* or nonspecific siRNA using Oligofectamine reagent (Invitrogen, CA, USA) according to the manufacturer's protocols.

2.6. cDNA microarray

Microarray System Analysis commissioned by Phalanx Biotech

Group Inc. First, a total of 1 µg of total RNA was amplified by Onearray Amino Allyl aRNA Amplification Kit. 5 µg Cy5-labeled aRNA was then hybridized on Human OneArray™ Chip (Agilent) and scanned by Agilent Microarray Scanner (G2505C). Image files were processed using Rosetta Resolver® System (Rosetta Biosoftware) to produce data files. After each group of microarray chips are normalized, the probe were chosen to have a significant difference if the $|\log_2(\text{Ratio})| > 1$ and $p < 0.05$. Genes that do not have a significant level ($p < 0.05$) will be excluded.

2.7. *In vitro* transfection

To create a stable DNMT1-expressing MiaPaCa2 cell line, MiaPaCa2 cells were transfected with 4 µg pcDNA3/Myc-DNMT1 (pDNMT1) plasmid (Plasmid #36939, Addgene, Inc., Cambridge, MA) [22], pcDNA3/Myc-DNMT3 A (pDNMT3 A) plasmid (Plasmid #35521, Addgene, Inc., Cambridge, MA) [23], or pcDNA3.0-neo vector by using Lipofectamine 2000 (Invitrogen). After 48 h, the cells were subjected to selection for stable integrants by exposure for 3 weeks to 500–800 µg/ml G418 in complete medium containing 10% FBS. The cells were then assessed for the overexpression of DNMT1 or DNMT3 A by using western blot analysis.

2.8. Western blot assay

The cell pellets were lysed with PRO-PREPTM protein extraction solution (INtRON Biotechnology, Korean) and incubated at -20°C for 20 min. Then centrifuged at 13,000 g for 5 min, and supernatant were collected. The concentration of the protein was measured by protein assay kit (Strong Biotech, Taipei, Taiwan). 30 µg protein were separated by 8% SDS-PAGE and transferred to PVDF membrane. The membrane was blocked with 5% nonfat milk, and then incubated with 1:1000 dilutions of primary antibodies for 16–18 h at 4°C . The primary antibodies were detected with horseradish peroxidase-conjugated anti-mouse or anti-rabbit as appropriate for 1 h at room temperature. HRP-conjugated secondary antibody was exposed with an ECL solution (Biological Industries) and detected by Chemiluminescence Imager (Fusion SoloS, Vilber, France) to visualize specifically labeled proteins. Then, the resulting exposed films were analyzed by densitometry.

2.9. DNA methyltransferase activity test

DNA methyltransferase activity was measured using the DNMT activity Quantification kit (abcam, Cambridge, UK), and the procedure was followed by instructions for the reagent group. After collecting the cells, the nuclei were separated and the proteins were extracted separately. Finally, 10 µg of nuclear protein was tested for activity. P84 antibody was used as an internal control.

2.10. Pyrosequencing

Bisulfite treatment and cleanup of DNA was performed by using Bisulfite Conversion Kit (Applied Biosystems, CA, USA). 2 µg genomic DNA was subjected to bisulfite conversion according to the manufacturer's instructions. In a subsequent PCR amplification of target region was performed by using PyroMark PCR Kit (Qiagen, Germany). The amplified region corresponds to Gene ID number 442213, nucleotides Chromosome 6, bp 48036204–48036223. The biotin-labeled PCR product was captured by Streptavidin-Sepharose HP (Amersham Pharmacia). PCR product bound on the bead were purified and made single stranded using a Pyrosequencing vacuum Prep Tool. The sequencing primers were annealed to the single-stranded PCR product, and pyrosequencing was used with PyroMark Q24 system (Qiagen, Germany). Quantitation of cytosine methylation was analyzed by the PyroMark Q24 Software.

2.11. Antitumor activity *in vivo*

All animal study was conducted in strict accordance with there commendations in the Guide for the Care and Use of Laboratory Animals of Tzu Chi University. The protocol was approved by the Institutional Animal Care and Use Committee of Tzu Chi University (approval number: #105-29 and #105-46). All surgery was performed under sodium pentobarbital anesthesia, and all efforts were made to minimize pain. Xenograft human tumor mice as a model to study antitumor activity of n-BP *in vivo*. Male immunodeficiency nude BALB/c (nu/nu) mice were purchased from National Laboratory Animal Center (Taipei, Taiwan). All experiments were carried out using 6 to 8 week mice. The animals were subcutaneously implanted with 1×10^6 MiaPaCa2 cells into the flank of mice. When the tumor volume reached 100 mm³, animals were randomly divided into three groups and treated with vehicle or 80, 500 mg/kg/day n-BP subcutaneous for five consecutive days (control group n = 4; treatment group n = 6). The mice were weighed two times a week up to days 28 and at same time, calipers measured the tumor volume. The tumor volume was calculated using the equation Length x Width x Height x 0.5236. Xenograft tumors were harvested and fixed in 4% formalin, embedded in paraffin for histologic study. For orthotopic tumor model, BALB/c (nu/nu) mice were injected with 2×10^6 GFP-tagged MiaPaCa2 cells orthotopically. After 29 days, tumor were detected by IVIS imaging system (IVIS Imaging System 200 Series, XENOGEN). Then, mice were divided into two groups and treated with control-wafer or BP-wafer (control-wafer group: n = 6; BP-wafer: n = 5). After 24 day treatment, tumor were detected by IVIS imaging system again. Mice were sacrificed when they met the end stage criteria of approved mouse protocol.

2.12. Preparation of CPP-SA copolymer containing BP (BP-wafer)

The synthesis of BP-polymer was described in the previous study [24]. In brief, the polymer were copolymer which were constituted by 1,3-bis (p-carboxyphenoxy) propane (CPP) and Sebacic Acid (SA). CPP-SA copolymers were synthesized by thermal melt polycondensation of CPP and SA in excess acetic anhydride. A mixture of CPP and SA prepolymers at a ratio of 20:80 was added to a glass bottle in a vacuum. The CPP and SA prepolymers were heated at 180°C in an oil bath for 1.5 h, and the pressure was reduced to 10^{-4} torr. Throughout the polymerisation, the vacuum pressure was reduced every 15 min. After 1.5 h, poly(CPP-SA) copolymers were washed with dichloromethane, and petroleum ether was then added to precipitate the poly(CPP-SA) copolymers. Finally, the poly(CPP-SA) copolymers were washed with anhydrous ether and dried in a vacuum. p(CPP:SA) (20:80) polymers containing BP were synthesized by high-pressure method and storage at -80°C avoiding light until experiment.

2.13. Release kinetics

BP-wafers were placed into a scintillation vial with 1.0 mL of 0.1 M PBS (pH 7.4) containing 1 mL n-octanol, and the mixture was incubated at 37°C . The solution was removed at various time points, and fresh buffer was added. The amount of BP released into the buffer was measured with a spectrophotometer. The O.D. at 310 nm determined BP concentration with methanol as background.

2.14. Immunohistochemical staining

Paraffin-embedded sections were prepared from the tumors and processed for immunohistochemical staining. Sections were deparaffinized in xylene and rehydrated using graded ethanol. Then there were incubated overnight in rabbit polyclonal anti-DNMT1 (1:100 dilutions; Novus, CO, USA), anti-Caspase 3 (1:500 dilutions; Abcam, Cambridge, UK), or anti-Ki67 antibody (1:100 dilutions; Abcam, Cambridge, UK) at 4°C . The immune complexes were visualized by the

HRP-DAB system (Thermo Scientific, MA, USA). Finally, sections were counterstained with hematoxylin, mounted, and observed under a light microscope (200×), and images were obtained.

2.15. Statistical analyses

The data for the experimental results were presented as mean ± SD. The data of the normal distribution were examined by Student's *t*-test analysis to statistics the significance of the experimental data. Survival analysis was presented by the Kaplan-Meier method. The *P* value of the statistical results < 0.05 is considered statistically significant.

3. Result

3.1. *n*-Butyridenephthalide could inhibit pancreatic ductal adenocarcinoma cell growth

The human DNMTs are often overexpressed in tumors and the reduction in DNA methylation levels may decrease the pancreatic tumor burden. To develop new DNMT inhibitors, first, we examined the expression of DNMTs protein in four pancreatic cancer cell lines (PANC1, MiaPaCa2, BxPC3, and HPAC) by western blot. The results showed MiaPaCa2 and BxPC3 had higher DNMT1 level and only MiaPaCa2 had a high DNMT3 A level (Fig. 1A). Therefore, we chose MiaPaCa2 to explore the mechanism in the following experiment because of MiaPaCa2 expressing the highest level of DNMT1, as well as DNMT3 A. Then using DNMT1 and DNMT3 A protein level as marker, we selected our drug database to search possible inhibitors of DNMTs. All compounds were treated MiaPaCa2 at same concentration for 24 h. DNMT1 and DNMT3 A were detected by Western Blot. The results showed that *n*-BP had the best inhibitory effect on DNMTs (Fig. 1B). Next, we tested whether *n*-BP had the ability to suppress PDAC cells. Using MTT assay, Human PDAC cells were treated with various concentrations of *n*-BP (concentration 0, 12.5, 25, 50, 75, and 100 µg/mL) for 24 h or 48 h. The results revealed that *n*-BP had a strong anti-proliferative effect on PDAC cells. The *n*-BP IC₅₀ value on MiaPaCa2, PANC1, BxPC3, and HPAC cells were 70 ± 3 µg/mL, 75 ± 7 µg/mL, 115 ± 4 µg/mL and 125 ± 5 µg/mL, respectively, after 24 h treatment (Fig. 1C, Supplementary Fig. 1).

3.2. *n*-Butyridenephthalide could induce apoptosis and cell cycle arrest

To determine the mechanism of *n*-BP causing growth inhibition, PDAC cells were analyzed with cell-cycle analysis. The results showed that *n*-BP treatment induced cell population arrest at the G0/G1 phase from 45.44% to 63.36% compared to control group (Fig. 1D). In addition, *n*-BP also led to a sub-G1 increase from 1.01% to 9.68%, and apoptosis related factor PARP activated in a dose-dependent manner (Fig. 1E). Those results indicate that the cells were in the process of cell-cycle arrest after *n*-BP treatment.

3.3. *n*-Butyridenephthalide could inhibit DNA methyltransferase family expression

In order to confirm the ability of DNMTs inhibition, the protein expression of DNMT family, including DNMT1, DNMT3 A, and DNMT3B, were analyzed with western blot. The results showed that the DNMT family was significantly decreased after *n*-BP treatment and decreased in a dose-dependent manner (Fig. 2A). Moreover, the protein expression level of DNMT family was suppressed in a time-dependent manner (Fig. 2B).

3.4. DNMT1 is required for mediating the inhibition of cell proliferation, but not DNMT3 A

In order to know how important the role of DNMTs in *n*-BP-induced tumor cell death, a vector system (pDNMT1) capable of sustained expression of DNMT1 was established in this study. MiaPaCa2 was transfected with the backbone vector or pDNMT1 plasmid, and was selected by G418. Western blot analysis indicated that exogenous DNMT1 recover DNMT1 level to parental type after *n*-BP treatment (Fig. 2C). In addition, in order to confirm that exogenous DNMT1 have the activity of methyltransferase, the total nuclear proteins were analyzed by DNMT activity assay. When cellular DNA methylation processes, S-adenosylmethionine, the principal methyl donor is converted to S-adenosylhomocysteine (SAH). The levels of SAH will be used as an indicator of cell methyltransferase ability. The results showed that compared with parental MiaPaCa2 treating with *n*-BP, the formation of SAH was recovered from 0.39 to 1.34 after *n*-BP 75 µg/mL treatment when the cells overexpressed DNMT1. (Fig. 2D). The cell viability was also evaluated by MTT assay. Human PDAC cells MiaPaCa2 were treated with various concentrations of *n*-BP for 24 h. We demonstrated that over-expression of DNMT1 could reduce *n*-BP cytotoxicity. In the presence of 75 µg/mL of *n*-BP, DNMT1 overexpression resulted in a significant increase by 24.4% in viability (Fig. 2E). In addition, MiaPaCa2 also was transfected with pDNMT3 A plasmid for create a stable DNMT3 A-expressing MiaPaCa2 cell line. However, exogenous DNMT3 A didn't suppressed *n*-BP cytotoxicity (Supplementary Fig. 2).

3.5. *n*-Butyridenephthalide induced protein instability of DNMT1

To demonstrate the mechanism of *n*-BP inhibiting DNMT1, the RNA level of *DNMT1* were analyzed in *n*-BP-treated PDAC cells. The results show *n*-BP did not alter the RNA level (Fig. 3A). We assumed *n*-BP influenced DNMT1 protein stability because the transcriptional level of *DNMT1* had not changed. In order to verify our hypothesis, CHX was used to block the protein synthesis. DNMT1 protein t1/2 was greater than 24 h in the CHX treatment only group. However, the DNMT1 protein had accelerated clearance in *n*-BP and CHX treatment group (Fig. 3B, C). In the two other cell lines, PANC1 and BxPC3, we also observed similar results. The combination treatment of *n*-BP and CHX rapidly degraded the DNMT1 protein. (Supplementary Fig. 3). Furthermore, the reduction of DNMT1 protein level by *n*-BP can be reversed by MG132, a proteasome inhibitor, suggesting that *n*-BP influenced the degradation of DNMT1 protein (Fig. 3D).

3.6. *n*-Butyridenephthalide rescued PTCHD4 expression via decreasing DNMT1

To gain insight into the mechanism by which DNMT1 regulated PDAC cell growth inhibition, we used cDNA microarray screening to identify *n*-BP-mediated changes in gene expression. Human PDAC tumor cells with pDNMT1 or pcDNA3 were treated with 75 µg/mL of *n*-BP. After 24 h incubation, RNA was extracted and used in microarray screening. We identified 21 potentially demethylated induced genes after *n*-BP treatment, including Patched domain containing 4 (PTCHD4), Apolipoprotein A-I, WAP four-disulfide core domain 6 (Table 1). In this study, we focused particularly on the characterization of *n*-BP-induced expression of *PTCHD4*. In order to confirm the microarray results, we examined the expression of *PTCHD4* in *n*-BP-treated MiaPaCa2 through qPCR and western blotting. *PTCHD4* mRNA and protein was significantly induced after *n*-BP treatment, in a dose-dependent manner (Fig. 4A, B). *PTCHD4* has been shown to be a negative regulator of canonical Hedgehog signaling [25]. To examine the effect of *n*-BP on Hedgehog signaling, SMO and GLI1 were analyzed by Western bolt. We found that SMO and GLI1 protein level were decreased after drug treatment in a dose-dependent manner (Fig. 4C). To determine whether *PTCHD4* is involved in *n*-BP-induced growth

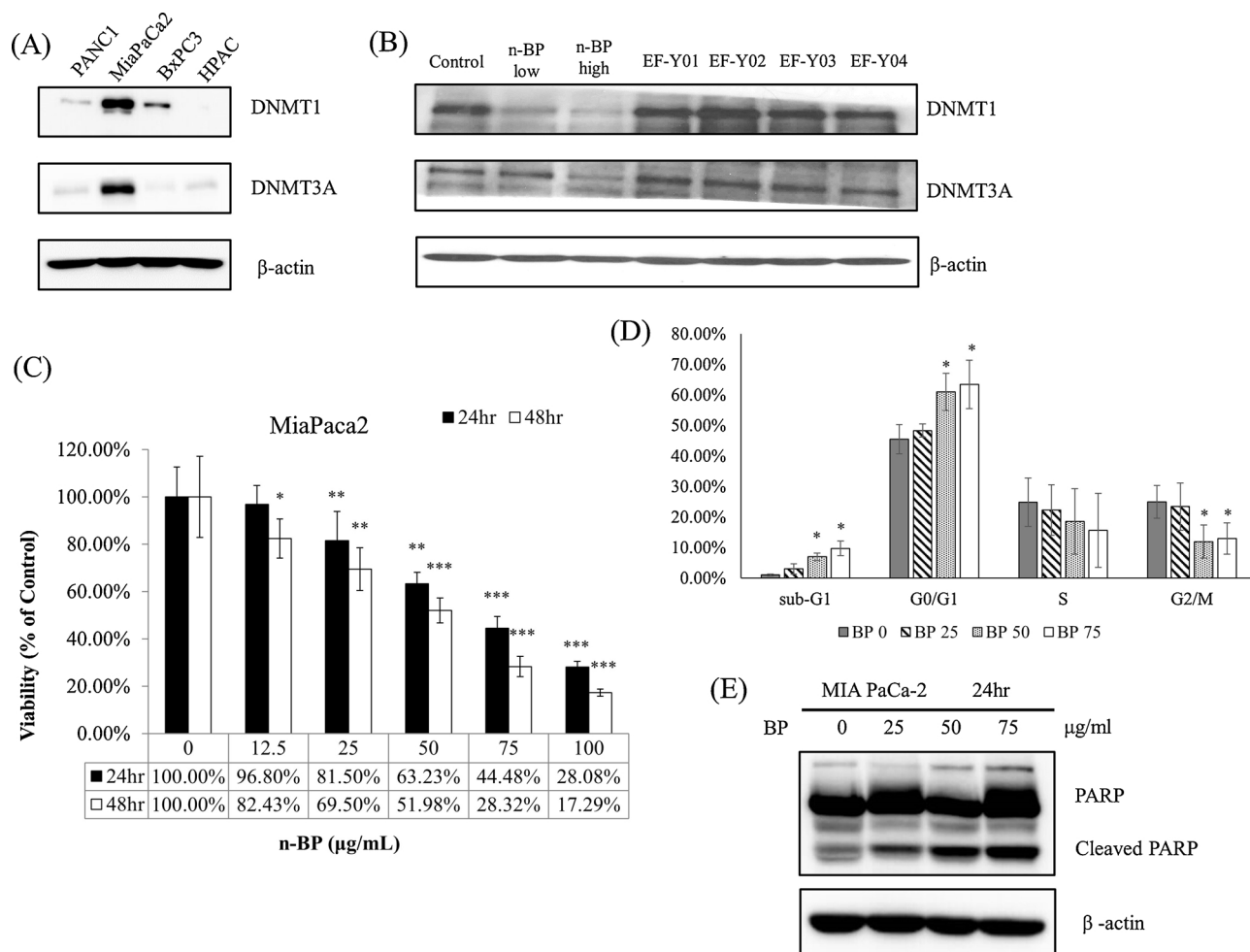


Fig. 1. n-BP induced cell cycle arrest and apoptosis in PDAC cells. (A) Western blot analyses of DNMTs in four PDAC cell lines. DNMTs were detected with β -actin serving as an internal control. (B) Analysis of different derivatives ability on DNMTs' protein inhibition. Western blot analyses of DNMTs' protein level. MiaPaCa2 cells were treated with different derivatives and its DNMTs' expression detected after 24 h treatment. (C) The cell viability was detected with MTT assay. MiaPaCa2 cells were seeded in a 96 well with a concentration of 8×10^3 /well. After 24 h, cells were treat with various concentration of n-BP for 24 or 48 h. After indicated time, cell viability was analyzed by MTT and normalized with control group. The data were presented as mean $\pm$ SD from three independent experiments. * : $p < 0.05$; ** : $p < 0.01$; *** : $p < 0.001$. (D) Propidium iodide staining for genomic DNA was used to analyze cell cycle. MiaPaCa2 cells were treat with various concentration of n-BP for 24 h. The data were presented as mean $\pm$ SD from three independent experiments. * : $p < 0.05$. (E) Apoptosis was detected by PARP activation. Cells were treated with n-BP for 24 h and analyzed by Western blotting. β -actin served as an internal control.

inhibition, *PTCHD4* siRNA were used to suppress *PTCHD4* expression. *PTCHD4* was suppressed in a concentration-dependent manner as confirmed by qRT-PCR (Fig. 4D). Then, the MTT assay was performed to identify cell viability after treatment with *PTCHD4* siRNA. We demonstrated that 29% and 33% of cell viability was rescued by 10 nM and 20 nM *PTCHD4* siRNA, respectively, after exposure of cells to 75 μ g/mL of n-BP (Fig. 4E). In addition, western blot analysis indicated that exogenous DNMT1 suppressed n-BP-induced inhibition of Hedgehog signaling (Fig. 4F). Moreover, using pyrosequencing confirms methylation ratio of two region *PTCHD4* promoters. We analyze 4 site of CpG islands upstream of the start site (Supplementary Fig. 4). The results showed that in control group the methylation ratio of site 1 to site 4 is 98%, 99%, 85%, and 100% respectively. After n-BP treatment, the methylation ratio decreased to 6%, 8%, 6% and 17% respectively. (Fig. 4G). These results indicated that n-BP may rescued *PTCHD4* expression and inhibited Hedgehog pathway through DNMT1 inhibition in MiaPaCa2 cells.

3.7. n-BP can inhibit tumor growth and reduce DNMT1 expression in a xenograft model

To evaluate the antitumor activity of n-BP *in vivo*, human PDAC cells were implanted through subcutaneous injection of MiaPaCa2 cells, into the flank of nude mice. After the tumor size reached 100 mm³, the mice were randomized into control and treatment groups, and administered daily subcutaneous injections of n-BP at 0, 80 or 500 mg/kg for five successive days, and then, tumor size was monitored for 28 days. The results show that the treatment with n-BP at 80 and 500 mg/kg reduced 53.5% and 80% tumor volume, respectively, compared with the control group (Fig. 5A). In addition, we didn't observe significant weight loss in animals treated with n-BP (Fig. 5B). The PDAC tumor tissues treated with n-BP exhibited downregulation of DNMT1 expression after treatment, as determined through immunohistochemical staining (Fig. 5C).

3.8. CPP-SA copolymer containing BP (BP-wafer) can inhibit tumor growth and prolong survival in a xenograft orthotopic model

In the past study, using a biodegradable CPP-SA polymer, we developed a local device, BP-wafer, to solve the current blood brain

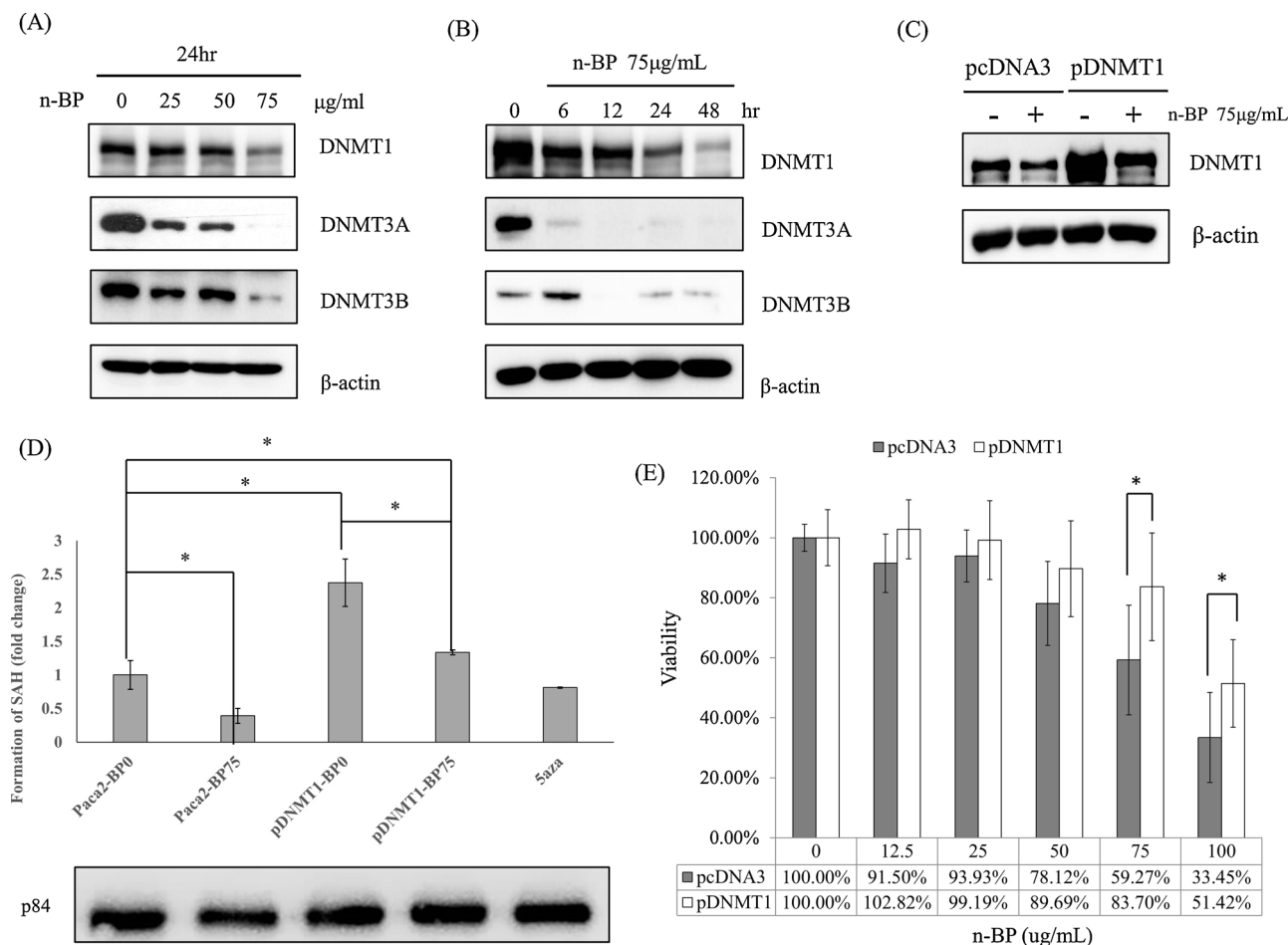


Fig. 2. n-BP reduce tumor growth mediated by DNMT1 inhibition in human PDAC cells. The western blot data demonstrate that n-BP can downregulate DNMT1, DNMT3A, and DNMT3B protein expression in MiaPaCa2 in tumor cells in a (A) dose-dependent and (B) time-dependent manner. β-actin served as an internal control. Overexpressed DNMT1 MiaPaCa2 cell line, MiaPaCa2-pDNMT1, was established. (C) DNMT1 overexpression was analyzed by Western blotting. (D) SAH formation was measured in the nuclear extracts prepared from MiaPaCa2 cell. MiaPaCa2 and MiaPaCa2-pDNMT1 were treated with n-BP for 24 h, and compared with an untreated control. 5-Aza-dC treatment was positive control. Protein immunoassay with p84 antibody was used as an internal control. (E) The cell viability was detected with MTT assay. MiaPaCa2-pcDNA3 and MiaPaCa2-pDNMT1 cells were treated with various concentration of n-BP for 24 h. The data were presented as mean ± SD from three independent experiments. * : $p < 0.05$; ** : $p < 0.01$.

barrier issue in glioblastoma [24,26]. It can clear residual tumor through side-by-side release. In order to determine whether BP-wafer could also work in PDAC, first, the release profile of BP-Wafers were measured *in vitro*. 17.5% BP-wafer released 50% of BP by 12 h and 25% BP-wafer released 50% of BP by 8 h. After 72 h release profile reached a plateau phase (Fig. 6A). Then, we tested the tumor, *in vitro*, the inhibiting ability of BP-wafer by membrane permeability. The results show that BP-wafer could release BP and reduce PDAC cell growth (Fig. 6B, C). The human PDAC orthotopic xenograft were established to evaluate BP-wafer efficacy *in vivo*. Immunodeficiency mice were injected with 2×10^6 MiaPaCa2 cells into the pancreas and BP-wafer was surgically implanted to the tumor site after 30 days of tumor injection. After 24 days of treatment, the tumor size was measured by IVIS (Fig. 6E). The Average of Radiance in 25% BP wafer ($5.87E + 07$ p/s/cm²/sr) was reduced by 58% compared with the control group ($1.41E + 08$ p/s/cm²/sr) (Fig. 6F). In addition, Kaplan–Meier survival curves was used to observe the mice survival rate. The results show the group of 25% BP wafer had a prolonged survival of 21.1 days compared with the control group ($p < 0.05$) (Fig. 6G).

4. Discussion

Aberrant CpG island hyper methylation, having been investigated in the early stages of PDAC and DNMTs protein expression, correlated

with advanced stages of the tumor [27,28]. Moreover, high DNMTs protein expression level of patients have been seen to have relatively poor survival [29]. In the past study, scientists have confirmed that depleted DNMTs could reduce the degree of methylation of the tumor and the tumor volume shrinkage [30]. At present, only two DNMT inhibitors have received approval from the U.S. Food and Drug Administration: 5-azacytidine (5-aza) in 2004 for the treatment of Myeloid dysplastic syndrome (MDS), acute myeloid leukemia, and chronic myelomonocytic leukemia (CMML); and 5-aza-2'-deoxycytidine (5-aza-dc) in 2006 for the treatment of MDS and AML respectively [31]. These two drugs belong to nucleoside-like molecules. The major mechanism is formation of an irreversible covalent bond with DNMT in enzyme catalyzed domain, so that DNMT cannot methylate DNA [13,32]. However, despite their high efficiency, their poor bioavailability, their instability in physiologic media and above all, their high toxicity restricted their use of both 5-aza and 5-aza-dc, which have strong side effects, including renal toxicity and bone marrow toxicity [33]. Hence, it is a great need for the development of effective and nontoxic inhibitors of DNMT. We scan a variety of nature compounds and found n-BP, was isolated from *Angelica sinensis*, can inhibited DNMT1 as well as DNMT3A.

The present study demonstrates that n-BP not only could reduce the expression of DNMTs protein in human pancreatic cancer cell lines with dosage and time, but also decreased tumor growth, with increasing n-

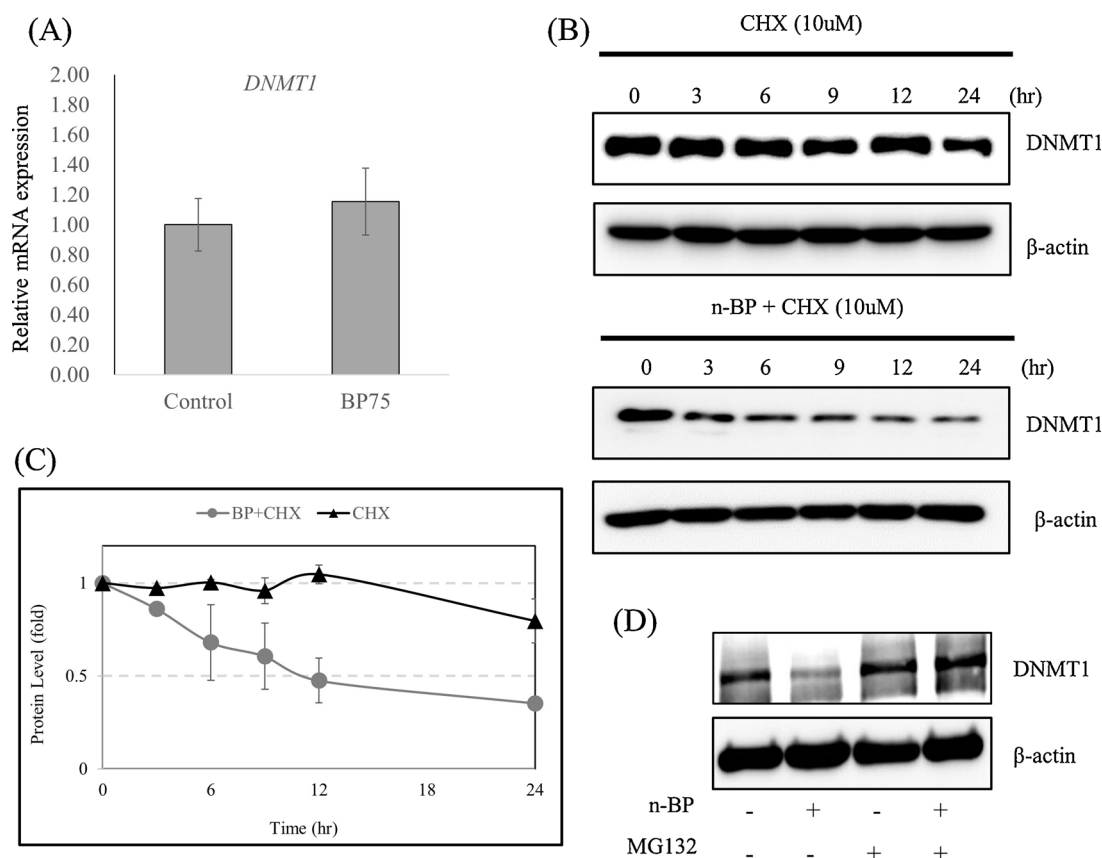


Fig. 3. n-BP regulate DNMT protein stability in human PDAC cells. (A) qRT-PCR was used to analyze *DNMT1* expression after n-BP treatment (B) MiaPaCa2 cells were treated with a combination of 10 μ M CHX and absence/presence of n-BP for the indicated time points. (C) DNMT1 half-life in the absence of n-BP treatment and n-BP treatment cells were analyzed. (D) MiaPaCa2 cells were treated with n-BP for 24 h in the absence or presence of 5 μ M MG132. The protein expressions were analyzed by Western blots. β -actin serving as an internal control.

BP dosage (Fig. 1). We hypothesized that DNMTs might be the target of n-BP against PDAC. In order to confirm the role of DNMT1 inhibition in BP-suppressing tumors, we used plasmid to overexpress DNMT1 with enzyme activity. The exogenous DNMT1 vector could recover DNMT1 level to parental type although DNMT1 expression in pDNMT1

expressing cell was still reduced upon n-BP treatment (Fig. 2). One possible explanation is that BP affects the stability of DNMT1 protein, and the stability of exogenous DNMT1 protein expressed by plasmid is still affected by n-BP. A large amount of exogenous DNMT1 could not 100% restore BP inhibitory DNMT1, but it did reduce the poisoning

Table 1

Accession No.	Representative Gene Name	Change fold#a	Change fold#b
NM_207499	Patched domain containing 4 (PTCHD4)	5.474	0.163
NM_000039	Apolipoprotein A-I (APOA1)	3.605	0.4
NM_080827	WAP four-disulfide core domain 6 (WFDC6)	3.593	0.288
NM_017839	Lysophosphatidylcholine acyltransferase 2 (LPCAT2)	3.479	0.337
NR_027336	Urate (hydroxyiso-) hydrolase	3.216	0.247
NM_020311	Atypical chemokine receptor 3 (ACKR3)	3.084	0.333
NM_031481	solute carrier family 25 (glutamate carrier), member 18 (SLC25 A18)	3.015	0.369
NM_001430	Endothelial PAS domain protein 1 (EPAS1)	2.959	0.384
NM_033294	Caspase 1, apoptosis-related cysteine peptidase (CASP1)	2.821	0.489
NM_207404	Zinc finger protein 662 (ZNF662)	2.704	0.442
NM_198475	Family with sequence similarity 171, member A2 (FAM171 A2)	2.656	0.42
NM_006205	Phosphodiesterase 6H (PDE6H)	2.564	0.434
NM_000575	Interleukin 1, alpha (IL1 A)	2.439	0.252
NM_015714	G0/G1 switch 2 (G0S2)	2.419	0.357
NM_004614	Thymidine kinase 2 (TK2)	2.394	0.406
NM_138396	Membrane associated ring-CH-type finger 9 (MARCH9)	2.37	0.466
NM_031442	Transmembrane protein 47 (TMEM47)	2.362	0.435
NM_000861	Histamine receptor H1 (HRH1)	2.344	0.479
NM_014009	Forkhead box P3 (FOXP3)	2.136	0.334
NR_045375	Uncharacterized LOC100506124 (LOC100506124)	2.117	0.487
NM_001164748	Ras association domain family member 8 (RASSF8)	2.111	0.307

#a: change fold between MiaPaCa2-pcDNA3 treated with BP and without BP.

#b: change fold between MiaPaCa2-pcDNA3 and MiaPaCa2-pDNMT1 treated with BP.

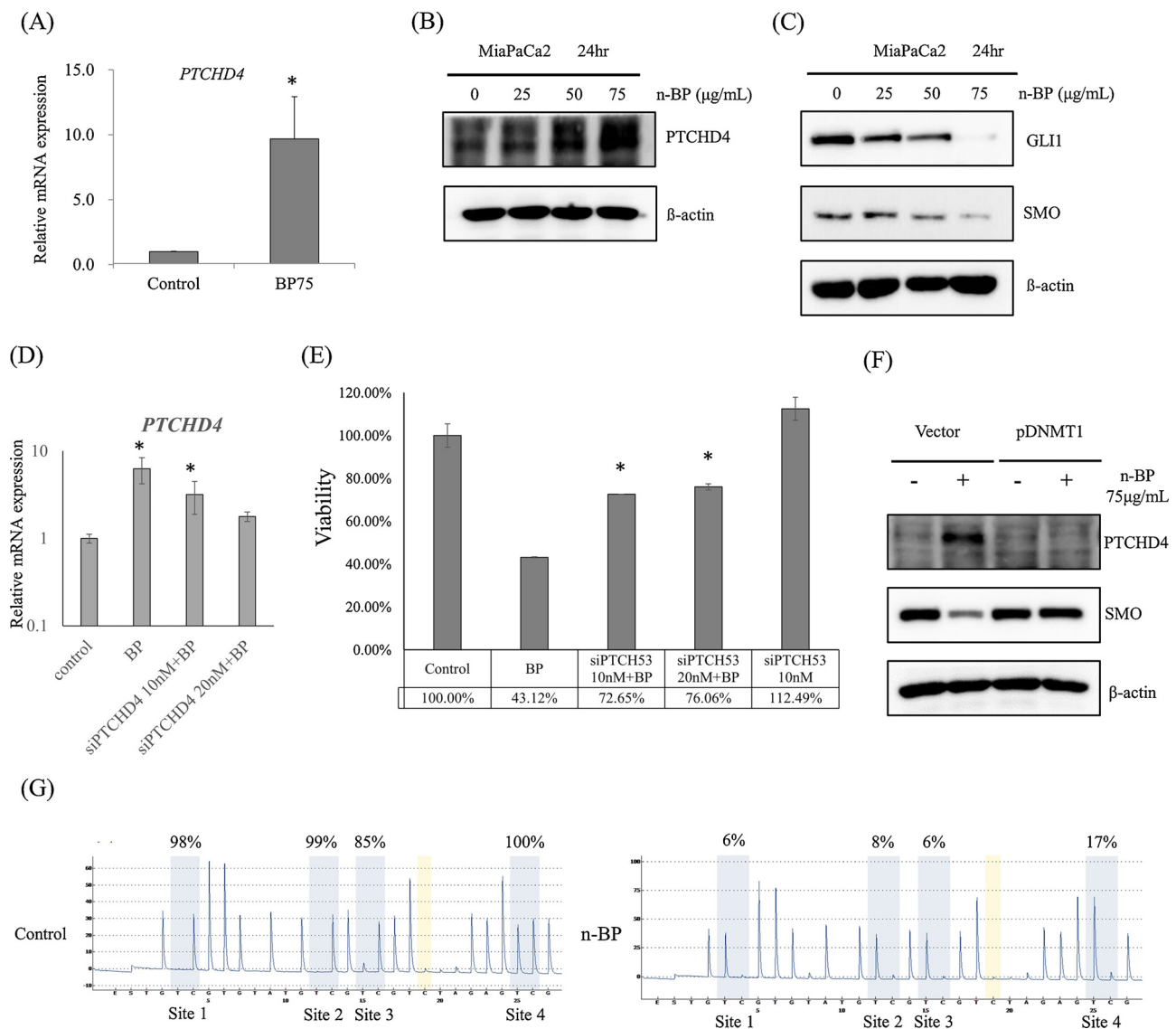


Fig. 4. n-BP regulates PTCHD4 expression and Hedgehog pathway in human PDAC cells. (A) qRT-PCR analysis demonstrates that n-BP can upregulate *PTCHD4* mRNA expression in MiaPaCa2. (B) The Western blot analysis demonstrate that n-BP can upregulate PTCHD4 protein expression in MiaPaCa2. β -actin served as an internal control. (C) Hedgehog pathway related-protein were detect by Western blot and β -actin served as an internal control. (D) MiaPaCa2 cells were transfected with *PTCHD4* siRNA (siPTCHD4) concentration of 10, and 20 nM as indicated for 48 h. After 48 h, *PTCHD4* expression level was analyzed by qRT-PCR after n-BP treatment. (E) Cells growth inhibition effect of n-BP was determined by MTT assay after siPTCHD4 transfection. (F) Hedgehog pathway related-protein were detect by Western blot after n-BP treatment in Overexpressed DNMT1 MiaPaCa2 cell line. (G) The region of the *PTCHD4* promoter examined for methylation by pyrosequencing.

ability of n-BP to PDAC cell, indicating that DNMT1 might be the target of n-BP. Another possibility is that, while compare endogenous DNMT1 was suppressed by n-BP in parental cancer cells, the DNMT1 expression at pDNMT1 was exogenous dominant. This result might implicate that n-BP can't inhibit exogenous DNMT1 expression. In contrast, in the overexpression of DNMT3A, viability was not restored by DNMT3 (Supplementary Fig. 2). According to these results, we proposed DNMT1, rather than DNMT3A, as a potential target of n-BP.

To clarify how n-BP affects DNMT1 protein expression, we analyzed the mRNA expression of DNMT1 and found that n-BP did not affect the mRNA expression of DNMT1 on a transcriptional level. While using the protein synthesis inhibitor CHX to treat PDAC cells with n-BP, we found that DNMT1 protein levels were decreased after treatment. In contrast, by using protease inhibitor to block the degradation of DNMT1, n-BP could not re-reduce DNMT1 protein (Fig. 3). Besides, protease inhibitor MG132 could rescue the n-BP regulated DNMT3A decreasing. However, MG132 could not stabilize DNMT3B. At present, we do not have a

clear explanation for the different results, and will continue in a further study (Supplementary Fig. 3). These results suggest that n-BP might inhibit PDAC tumor growth, mediated by regulation of protein degradation of DNMT1 rather than its transcriptional level. DNMT inhibitors are divided into two families: the nucleoside analogs whose mechanism rely on DNA incorporation, and the non-nucleoside inhibitors which structure varies according to their inhibitory mechanism [31]. Recently, a non-nucleoside-like compound SGI-1027, a lipophilic quinoline was revealed to be a novel DNMT inhibitor. Although the mechanism is unclear, it has been observed a rapid proteasomal degradation of DNMT1 after treatment of colon cancer cells by SGI-1027 [34]. In this study, we demonstrated that n-BP also causes the degradation of DNMT1. Hence, we consider n-BP may have similar mechanism to inhibit DNMT1.

In 2011, Clinic immunohistochemistry analyses by Wang et al. suggested that the expressions of Hedgehog pathway related gene GLI1 and DNMT1 were highly correlated in pancreatic cancer tissues [35].

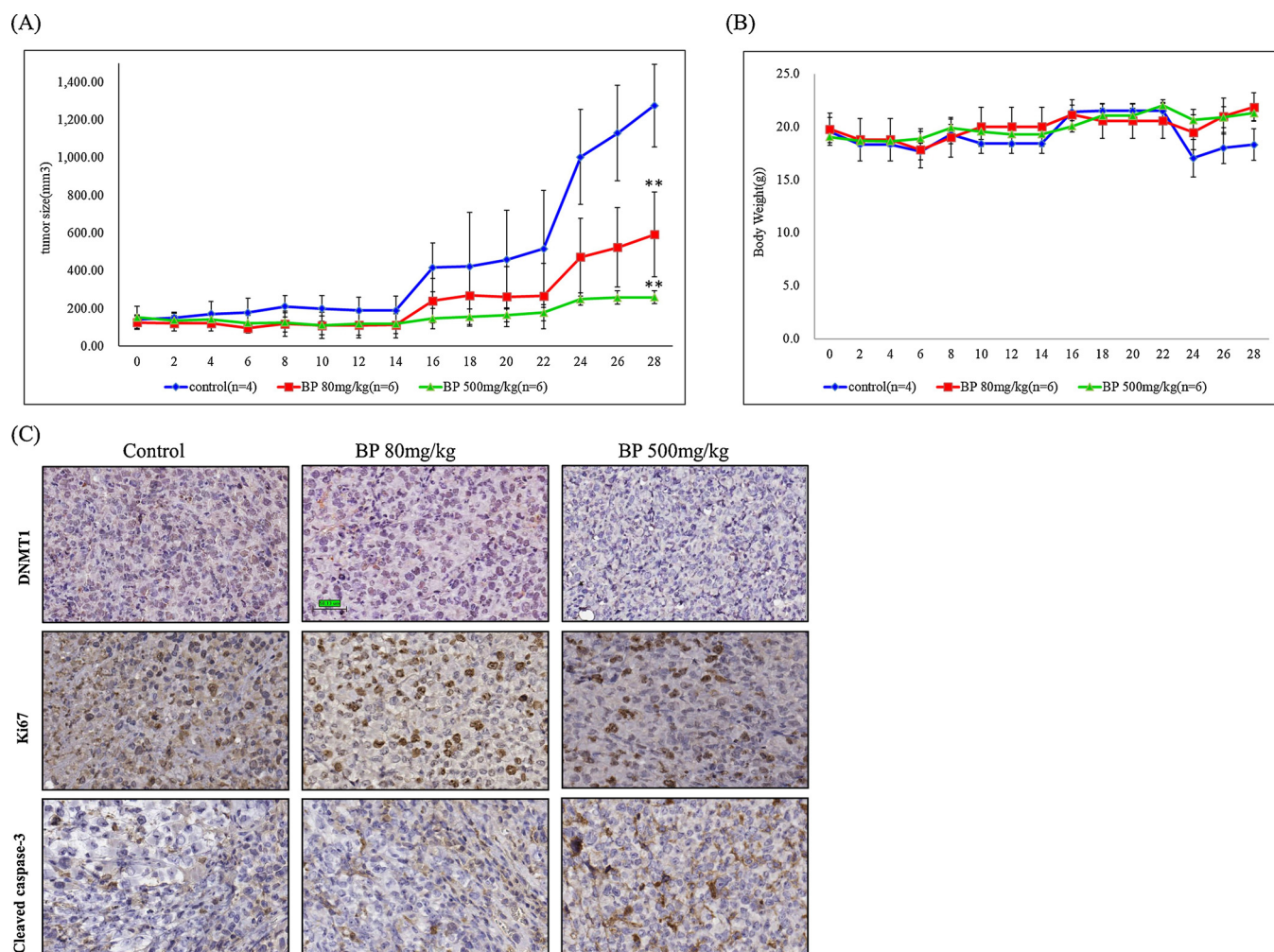


Fig. 5. n-BP could reduce tumor growth and suppresses DNMT1 expression in MiaPaCa2 xenograft tumors in nude mice. Human xenografts tumor model were established by subcutaneous injection 2×10^6 MiaPaCa2 cells into the flank of nude mice. After tumors reached approximately 100 mm³, mice were given daily subcutaneous injections of vehicle alone or the indicated amount of n-BP for 5 successive days (control group n = 4; treatment group n = 6). (A) The volumes of the tumors and (B) mice body weights were measured twice a week. The data were presented as mean $\pm$ SD. * $p < 0.05$. (C) Expressions of Caspase 3, Ki67, and DNMT1 were analyzed by immunohistochemical staining in MiaPaCa2 xenografts tumor.

However, it is still unclear whether DNMT regulated GLI1 and Hedgehog signaling in PDAC. In this study, using cDNA microarray, we demonstrated that *PTCHD4*, a regulator of canonical Hedgehog signaling, was upregulated after n-BP treatment while DNMT reduced. This result proposes that n-BP could decrease Hedgehog pathway-induced tumor proliferation via DNMT1 inhibition. Using western blot, we demonstrated n-BP decreased GLI1 and SMO expression in a dose-dependent manner (Fig. 4). Meanwhile, *PTCHD4* was inhibited by *PTCHD4* siRNA, and PDAC cells viability was restored after n-BP treatment (Fig. 4). Further, we also confirmed that the *PTCHD4* promoter methylation would be decrease after n-BP treatment. In summary, these results showed that n-BP inhibited DNMT and its downstream Hedgehog/*PTCHD4*.

Previously, we found that n-BP suppressed GBM through multi-target including orphan nuclear receptor *Nur77*, receptor tyrosine kinase *AXL*, and S-phase kinase-associated protein 2 [26,36,37]. In this study, we investigated how n-BP could inhibit DNMT1 and decrease PDAC cells growth. However, after overcoming the DNMT1 inhibition, viability of PDAC cells did not restore completely. These results suggest that n-BP doesn't just target DNMT1. We additionally demonstrated that n-BP could also induce the orphan nuclear receptor in PDAC cells (data not shown). In the next step, we will confirmed whether the orphan nuclear receptors are another target of n-BP inhibiting PDAC.

After dermal application most of the n-BP can permeate through the skin into the peripheral circulatory system within 48 h without accumulation in the skin; distributes to lung, liver, bile and kidney; and then is excreted into urine as a cysteine conjugate. On the other hand, after intravenous administration n-BP, 70% of the administered n-BP was excreted within 8 h and about 80% within 24 h, while only 5% was excreted into feces within 24 h [38]. In our experiment, MiaPaCa2 cells (1×10^6) were transplanted into the backs of nude mice and treated with n-BP through subcutaneous injection. The dose of n-BP we refer to the previous experiment and choose the minimum effect dose and maximum tolerated dose in GBM. After treatment with n-BP at 80 and 500 mg/kg reduced 53.5% and 80% tumor volume, respectively, compared with the control group (Fig. 5A). In addition, the body weight did not significantly change in the observed days (Fig. 5B). Previous study also showed that BP does not cause drug-related toxicity in major organs [39].

Over the past four decades, a variety of systemic chemotherapy was unable to effectively extend the survival rate of patients with pancreatic cancer, with five-year survival rates of less than 8% [40]. The main reason is that PDAC is often accompanied with the formation of highly fibrous tissues due to abnormal activation and proliferation of stellate cells around the tumor. These fibrosis tissues resulted in vascular deficiencies and hypoxia microenvironment and blocked systemic

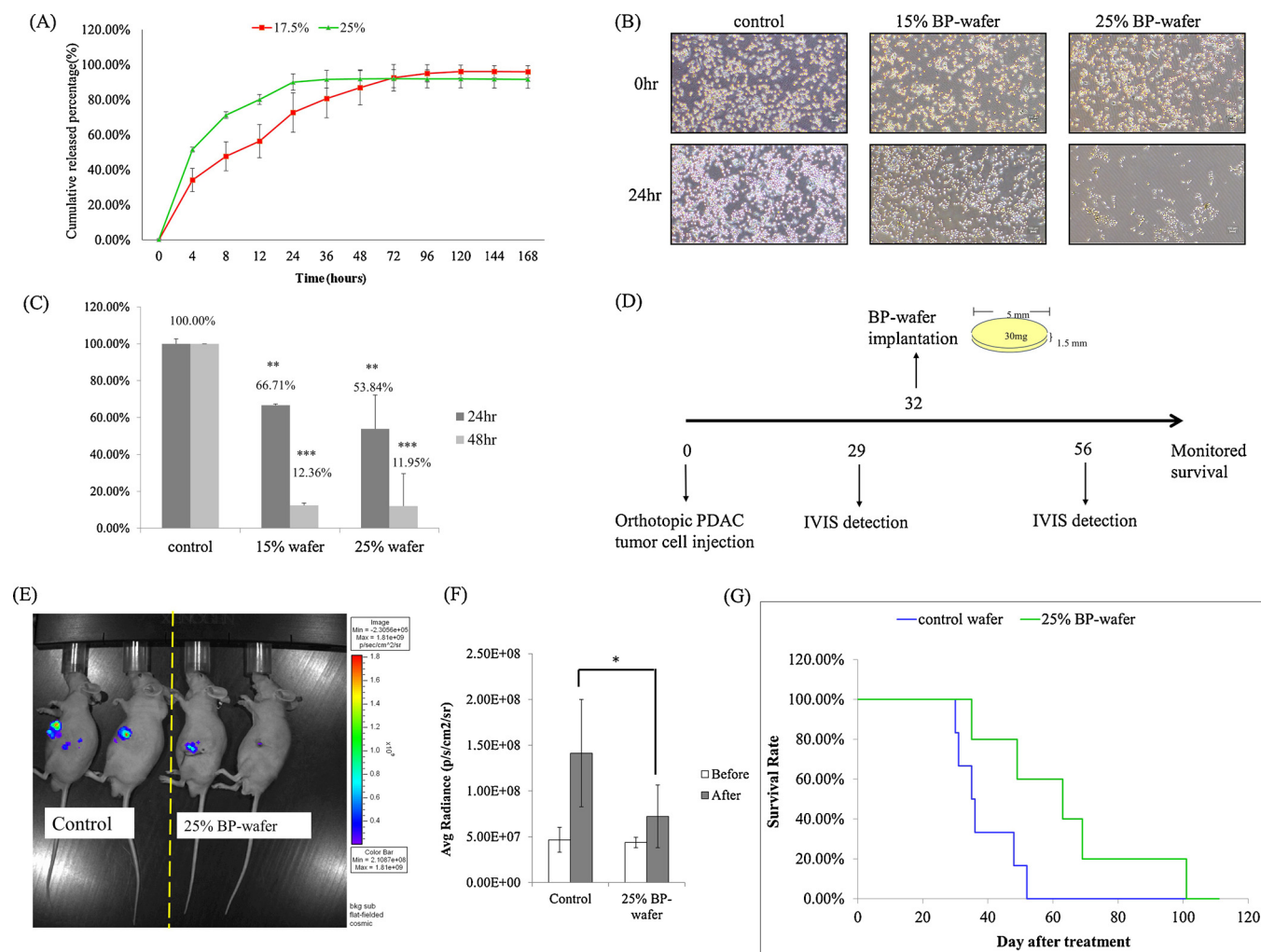


Fig. 6. BP-wafer could against human PDAC cells *in vitro* and *in vivo*. (A) The release kinetics of 17.5% and 25% BP-wafers measured over 168 h *in vitro*. Sustained release of BP was observed with both wafer formulations. The human PDAC MiaPaCa2 cells were seeded in membrane permeable well and treated with control wafers, 15% BP-wafer, and 25% BP-wafer for 24 h or 48 h. (B) Cell morphology of MiaPaCa2 cells were shown. (C) Cell viability was determined by MTT assay. The data are expressed as mean $\pm$ SD. ** : $p < 0.01$; *** : $p < 0.001$. (D) The immunodeficiency mice were orthotopically implanted with GFP-tagged MiaPaCa2 (2×10^6) and treated with control-wafer or BP-wafer (control-wafer group: $n = 6$; BP-wafer: $n = 5$). After 24 days of treatment, (E) the tumor volume was detected by non-invasive live-imaging system (IVIS). (F) The tumor volume was quantified by average of radiance (p/s/cm²/sr). The data are expressed as mean $\pm$ SD. * $p < 0.05$. (G) Mice survival was monitored and using Kaplan-Meier survival curves analysis, the relative survival rate is shown for each group (* $p < 0.05$).

chemotherapy penetration [41,42]. Therefore, improving the drug penetration and reaching the tumor site is the first priority in order to treat PDAC. The Nanoliposomal irinotecan, which is a FDA-approved compound in the treatment of pancreatic cancer in 2015, is incorporating irinotecan with niacin liposomes. However, the median survival rate increased by only 1.9 months in clinical patients because of the reliance on systemic delivery. Even the new delivery enhanced penetrating ability [43]. In addition, in order to address the difficulty of drug entry into pancreatic cancer cells, Indolfi et al. developed a local drug delivery device, paclitaxel eluting device (PED), which used Poly (lactic-co-glycolic acid) (PLGA) to carry Paclitaxel [44]. The PED is able to linearly release high doses of chemotherapeutic drugs for up to 60 days and at least a 2-fold reduction of viable tumor volume in orthotopic xenograft model in mice. The PED suggested that local drug delivery platform could effectively overcome intrinsic and extrinsic PDAC chemoresistance, mediated by tubulin polymerization. However, it is not tumor specific and presently has a lot of side effects.

In our study, after oral administration in rats, z-BP biodistribution was determined by liquid chromatography-mass spectrometry/mass spectrometry (LC-MS/MS) method. The biodistribution data showed that z-BP entered spleen, pancreas, small intestine, and spinal cord. The

pancreas concentration is second the highest organ. In the past decade, we developed a device, BP-wafer that could solve the current blood brain barrier issue in glioblastoma [24]. Using a biodegradable CPP-SA polymer, BP-wafer, that can clear the residual tumor through local release in order to reach higher intratumoral concentrations not achievable by the systemic administration; the phase I/IIa clinical trial has been approved by FDA now. In this study, we conducted an animal model of human pancreatic cancer in nude mice, and the inhibitory effect of BP-wafer on human pancreatic cancer, which was analyzed. We demonstrated the mean optical signal intensity in BP-wafer reduced to about 58% compared with the control group (Fig. 6). The mean survival time was found to be 43.29 days in the control group and 64.40 days in the treatment group, which significantly prolonged survival rate by 21.11 days ($p < 0.05$) (Fig. 6). The reasons include that not only did n-BP target DNMT/PTCH, but it could diffused far distances, leading to unresolvable infiltration tumor cell apoptosis [26]. In addition, since the pancreatic cancer tends to attack the surrounding tissue, Indolfi et al. also found that a local device can be used as a physical barrier to help control the local infiltration of the tumor [44]. Together, these results suggest that local biodegradable polymer loading with DNMT1 inhibitor n-BP might be useful for targeted cancer therapy.

5. Conclusions

Our study showed that a novel potential DNMTs inhibitor, n-BP could inhibit tumor growth by downregulating DNMT1. In addition, a biodegradable interstitial release polymer loading BP exhibited an extended diffusion distance, increasing local BP concentration, prolonged release time, and significantly prolonging animal survival. This may contribute to a potential therapeutic target for clinical intervention.

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Conflict of interest

The authors declare no conflict of interest.

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:<https://doi.org/10.1016/j.phrs.2018.10.015>.

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